PROJECT RAKSHAK 2.0

Technical Guidelines, KPI Explainer & Judge Review Guide

Bullet-point edition • Fully updated to present project state

PROJECT SNAPSHOT	SOURCE-DOCUMENT DETAILS
Hackathon	KLS Hackfest 2026 — Problem Statement 1A (Naval/Army Geospatial Threat Detection)
Team	Team BotS — KLS Gogte Institute of Technology / KLE Technological University
System	Sovereign dual-domain C4ISR and geospatial threat-intelligence prototype
Core idea	Fuse satellite imagery, SAR/AIS correlations, drone EO/IR, ground sensors and tactical workflows in an offline-first Common Operating Picture.
Deployment & Remote	Air-gapped FastAPI backend + React/Leaflet UI, Dockerized, verified clean on judge repo: HackFest-2026/Team1A.2
Audit & Tests	12/12 Platform Smoke Tests PASSED • 38/38 Pytest Tests PASSED • 0 Outbound Sockets (Zero-Egress Verified)

Important: figures in this guide are reported directly from canonical evaluation JSON reports in evaluation/results/, evaluation/RESULTS.md, and PROJECT_OVERVIEW.md. “Measured”, “simulated”, “computed”, “assumed”, “projected” and “not done” are kept strictly distinct. This updated edition incorporates the zero-leakage split audit, reconciled vessel mAP benchmarks, standardized Jetson dense-FP16 roofline intervals, 3 distinct offline map styles, and interactive Mission Planner execution.

How to use this guide

- Use the short “Judge explanation” under each KPI as a plain-language answer during judging.
- Use the evidence bullets to explain what was actually tested and on what hardware/data.
- Never describe a simulated or projected result as a live field result.
- Where the source marks a test “NOT DONE”, say so directly and describe it as future validation.
- Refer to evaluation/RESULTS.md and underlying JSON reports for exact audit provenance.

Project in one minute

- **Problem:** dark vessels can evade AIS; analysts must inspect large imagery volumes; cloud dependence is unsuitable for disconnected or radio-silent operations.
- **Solution:** an offline-first, multi-modal COP that combines optical detections, SAR/AIS correlation logic, temporal change detection, drone EO/IR, unattended ground sensors and mission-planning support.
- **Security goal:** zero external data egress during the audited application-level test; local services communicate strictly over loopback.
- **Operational value:** consolidate duplicate detections, prioritize alerts, retain queued alerts through simulated link outages and help analysts move from raw sensor data to a reviewable operational picture.
- **Prototype boundary:** functional, grounded prototype with measured benchmarks and simulated scenarios. Not yet validated by physical sea trials, military radio trials, independent GPS surveys or a human military-staff exercise.

Status vocabulary

Label	Meaning for judges
MEASURED	Timed or evaluated in the stated test environment; does not automatically mean field-validated.
SIMULATED	Produced by a controlled synthetic or Monte Carlo scenario; useful for algorithm testing, not proof of live operational performance.
COMPUTED	Calculated from measured outputs and/or stated assumptions.
ASSUMED	Depends on a modelled baseline or a human-time assumption rather than an observed comparative trial.
PROJECTED	Estimated for hardware or a full pipeline not actually benchmarked end to end.
NOT DONE	No result is available; do not imply validation.

KPI 1 — Optical detection quality: precision, recall and mAP

- Headline overall YOLO11m FP16 results on the held-out validation set: **mAP@50 48.46%**; mAP@50–95 14.45%; precision 53.78%; recall 48.17%.
- Test scope: 540 tiles, 38 scenes and 66,521 labelled instances. A separate clean maritime holdout contains 21 tiles and 280 vessels across 5 scenes, with 0 training overlap verified.
- Aircraft perform best: **mAP@50 89.85%**, recall 86.67% (45 ground-truth aircraft). Vehicles: mAP@50 46.83%, recall 52.61%. Infrastructure: mAP@50 44.41%, recall 38.75%.
- Vessels remain the key weakness: generalist mAP@50 **12.75%** and recall 14.65%; the YOLO11n 1024px specialist reaches mAP@50 **16.24%** and recall 23.57% on the maritime holdout.
- FP16 overall mAP@50 (48.46%) is close to FP32 (48.37%) and ONNX CPU (47.80%) in the reported tests.
- **Zero-Leakage Split Audit:** split_overlap_audit.json audited 276 specialist scenes, 301 generalist scenes, 37 val_tune scenes, and 38 val_report scenes. Result: **ZERO overlap across all evaluation pairs** (disjoint holdout confirmed).
- **Reconciled Vessel Numbers:** Generalist vessel mAP appears as 12.75% (540 tiles holdout), 12.07% (21 tiles edge bench), and 12.26% (21 tiles boost baseline); 12.75% and 16.24% (specialist) remain the authoritative headline values.

Judge explanation: “Precision tells us how often a detection is correct; recall tells us how many real labelled objects we find. mAP summarizes detection quality across confidence thresholds and box overlap requirements. Our model is stronger for aircraft than small vessels, so we report that limitation honestly rather than presenting one overall score as universal accuracy.”

Be ready to clarify: mAP@50 allows a looser box-overlap threshold than mAP@50–95. Low vessel scores are consistent with small 12–35 pixel objects and class imbalance reported in the source.

KPI 2 — Dark-vessel detection

- Headline end-to-end capture: **79.40% ± 2.80%**; 95% confidence interval 78.09%–80.71%.
- Post-detection recall: **92.63% ± 2.23%**; CA-CFAR radar detection rate: **85.73% ± 2.65%**.
- Reported simulated precision: 100%; false-dark alarm rate: 0%; F1: 0.9616 ± 0.0120.
- Evaluation uses 20 random seeds and 10,000 synthetic contacts per sweep condition; 30% of the synthetic contacts are configured as dark vessels.
- Dead-reckoning matters under timestamp drift: without propagation, false-dark alarms rise to 15.20% at ±5 minutes, 61.08% at ±15 minutes and 77.45% at ±30 minutes in the simulated scenarios.
- Physical sea-trial ground truth: NOT DONE; the source states no local xView3-SAR dataset was present in the repository.

Judge explanation: “AIS is a cooperative broadcast, so a vessel that stops transmitting may disappear from normal AIS tracking. This KPI tests whether radar-like SAR detections can be correlated with expected AIS positions to flag possible dark vessels. The encouraging numbers are simulation results, not a claim of proven live sea performance.”

KPI 3 — Sensor-to-alert latency

- YOLO11m single-tile inference on an NVIDIA RTX 4060 Laptop GPU: 34.9 ms mean unboosted; **17.85 ± 0.11 ms** in the boosted benchmark (P50: 17.37 ms).
- Boosted inference throughput: **55.65–56.50 frames/tiles per second** for the stated 1024×1024 model configuration.
- Dual-engine YOLO11m + YOLO11n with Weighted Box Fusion: 47.9 ms mean and 20.87 tiles/second.
- Full-scene ingestion across 38 scenes / 540 tiles: mean 1.62 s, median 1.58 s, P95 2.30 s in the reported boosted setup.
- Full end-to-end target below 1.8 s is marked PROJECTED; the source also reports P95 2.30 s and an unboosted estimate around 3.1 s.
- SAR CA-CFAR processing (1.1 s) and SAR-to-AIS matching (85 ms) are described as code-path timings, not logged benchmark measurements.

Judge explanation: “Latency is the time the system takes to process input; throughput is how many inputs it can process per second. The fast GPU number is model inference, not the whole journey from raw sensor feed to an alert. We therefore separate tile timing, full-scene ingestion and the projected full decision pipeline.”

KPI 4 — False alarms and alert pacing

- Precision-oriented threshold mode: **9.93 unmatched false positives per km²** at IoU ≥ 0.3, with 94.75% precision but 17.05% recall.
- F1-oriented mode: 196.83 false positives/km² at IoU ≥ 0.3, with 76.90% precision and 62.36% recall.
- Older thresholds report 49.78 FP/km², 88.59% precision and 36.77% recall at IoU ≥ 0.3.
- At 12 scenes/hour surveillance pacing, the report computes 7.26 high-threat alerts/hour and 149.37 medium-threat alerts/hour; high + medium combined is 156.63/hour.
- 24-hour continuous live-feed false-alarm testing and physical sea-spray/wave-clutter tests: NOT DONE.

Judge explanation: “This is a trade-off. Raising confidence thresholds reduces false alarms but misses more real objects; lowering thresholds catches more possible threats but asks analysts to review more false positives. We can tune the operating point to the mission, but we have not claimed that one setting solves every deployment.”

KPI 5 — Track continuity and ID switches

- Tracking combines detections with a constant-velocity Kalman filter in a local ENU coordinate frame and data-association methods including Hungarian, JPDA and nearest-neighbour.
- Clean simulated control: **0 ID switches**; MOTA 96.08%; MOTP 2.27 m; 96.08% track continuity.

KPI 5 — Track continuity and ID switches (cont.)

- Crossing paths: **0 ID switches** in the stated test; MOTA 92.16%, with MOTP around 9.56–9.99 m.
- During a simulated 60-second AIS dropout: **0 ID switches** and JPDA MOTA 93.44%.
- Harder chokepoint scenario: JPDA reports 2 ID switches and MOTA 71.22%; Hungarian reports 0 ID switches but MOTA 51.22%.
- Physical field-radar multi-target trials: NOT DONE.

Judge explanation: “A track ID should keep referring to the same object over time. An ID switch means the tracker accidentally swaps identities. MOTA summarizes missed detections, false detections and identity errors; MOTP describes position-match precision. Results are from simulated trajectories, so live radar validation remains future work.”

KPI 6 — Geolocation error (CEP)

- Overall, for matched detections at IoU ≥ 0.3: **CEP50 0.70 m**, **CEP90 2.43 m**, mean error 1.17 m and RMSE 2.03 m.
- At IoU ≥ 0.3, 37,300 of 66,521 ground-truth objects (56.07%) had matched predictions.
- Vessel subset: CEP50 0.92 m and CEP90 6.54 m, but only 78 of 280 vessels (27.86%) were matched.
- CEP50 is the radius containing 50% of matched localisation errors; CEP90 is the radius containing 90%.
- Important qualification: error is conditional on a bounding-box match and measured against the dataset’s GeoTIFF georeferencing metadata, not independent GPS ground truth.
- Independent differential-GPS survey: NOT DONE.

Judge explanation: “This KPI asks how close the map coordinate is to the dataset’s georeferenced location. CEP50 of 0.70 m means half of the matched predictions are within that radius. It does not mean every target is located within 0.70 m, and it does not measure missed targets. The reference is GeoTIFF metadata, not a field GPS survey.”

KPI 7 — Edge deployment, model size and power

- YOLO11m generalist: 20.1 million parameters; FP16 checkpoint 38.7 MB; ONNX export 40.5 MB.
- YOLO11n vessel specialist: 2.6 million parameters (1024px fine-tuned).
- On the RTX 4060 laptop GPU, boosted inference reaches 55.65–56.50 FPS; measured GPU board power during FP16 inference is about 43 W gross and about 30.4–30.8 W net over 12.64 W idle baseline.
- **Standardized Dense-FP16 Jetson Roofline Intervals (basis 35.74 ms host, no TensorRT factor):**
- **Jetson AGX Orin 64GB: 44.7–48.8 ms** (**20.5–22.4 FPS**) [unmeasured, rough estimate only; basis 35.74 ms host].
- **Jetson Orin Nano 8GB: 134.6–203.1 ms** (**4.9–7.4 FPS**) [unmeasured, rough estimate only; basis 35.74 ms host].
- *TensorRT may improve this; unmeasured, no Jetson run.* All earlier single-point estimates (19.4 ms, 38.2 ms) and 10.8–23.8 ms / 42.0–92.6 FPS intervals are formally superseded.
- All Jetson values are rough projections, not physical measurements. Jetson power measurement is NOT DONE; physical Jetson hardware was not available.

Judge explanation: “Edge readiness means the model is small and fast enough to be a candidate for local hardware, without relying on cloud inference. We measured performance on an RTX 4060 laptop. The Jetson numbers are roofline scaling estimates only; we would need to run the actual model on those boards and measure power before claiming validated embedded performance.”

KPI 8 — Analyst workload reduction and auto-triage

- 14,008 raw detections were consolidated into 2,379 spatial entities: **83.02% fewer review items** after 500m clustering.
- After auto-closing 493 isolated civilian-vehicle entities, 1,886 items remain for human review: **86.54% fewer review items** than reviewing every raw detection.
- Auto-close rate: 20.72% of consolidated entities. **100% of all 493 auto-closed items were isolated civilian vehicles**.
- Infrastructure is never auto-closed; vessels and aircraft are routed directly to the priority queue.
- Safety Suite v2: 9/9 critical threat cases escalated (0 missed); 4/4 cooperative vessel controls safely auto-closed; 2/2 ambiguous cases routed to human review.

KPI 8 — Analyst workload reduction (cont. — Workload baselines)

- **Baseline A (Manual Raw Detection Review):** Reviewing 14,008 raw detections vs 1,886 items yields a measured **86.54% reduction** across 45–240 s review assumptions (e.g. saving 47.85 hrs/surveillance hour at 45 s/item).
- **Baseline B (Fixed 40 min/scene budget = 8.0 hrs/hr cap):** Reviewing 595.6 items/hr at ≥60 s requires 9.93h–39.71h, exceeding the fixed 8.0h cap. This explains why rows at 60 s (-24.1%), 120 s (-148.1%), and 240 s (-396.4%) show negative differentials relative to this artificial ceiling.
- Workload-hour savings are COMPUTED using assumed review durations. They are not the result of a human analyst user study.

Judge explanation: “The key idea is to reduce duplicate boxes and group them into entities before an analyst reviews them. The 86.54% figure is an item-count reduction, not a measured claim that every analyst will work 86.54% faster. The safety rules keep vessels, aircraft and infrastructure out of automatic closure; humans remain in the decision loop.”

KPI 9 — Mission-planning cycle and SITREP

- Across 30 runs of three tactical defense scenarios, average local REST machine time is **0.1843 seconds** (Scenario A: 0.257s, Scenario B: 0.140s, Scenario C: 0.156s).
- Operator-paced estimate averages **50.19 seconds**, including an assumed 50 seconds of operator thinking time.
- Manual baseline averages **24 minutes (1440 s)**, modelled from a standard five-step military staff procedure.
- Computed cycle-time reduction is **96.51% (or 96.5%)** using the operator-paced framing, and **99.98%** for machine REST execution.
- STANAG 2014 SITREP compilation is reported below 50 ms; local BM25 doctrine retrieval below 40 ms.
- Human military-staff exercise and real tactical radio order dissemination: NOT DONE.

Judge explanation: “The services can assemble a local response quickly, but the headline reduction compares software timings with a modelled manual workflow. We have not run a controlled study with military staff, so we present 96.51% as a computed estimate rather than observed human performance.”

KPI 10 — DDIL availability and alert survivability

- DDIL means denied, disrupted, intermittent or limited communications.
- Simulated 30-minute mission: **100% edge detection uptime** over 720 alert epochs; 720/720 alerts delivered after store-and-forward; **0 lost alerts**.
- 13 duplicate packets were handled by sequence-numbered idempotency logic without duplicate ingestion.
- After a simulated 300-second blackout, resynchronisation averages 40.89 seconds at 32 kbps with packet loss; 10.06 seconds at 512 kbps without packet loss.
- Picture availability is 60.36% under a ≤10-second staleness definition, 68.90% under ≤30 seconds and 72.38% under ≤60 seconds.
- Physical combat-net radio trials: NOT DONE.

Judge explanation: “The prototype queues alerts locally when connectivity is lost and forwards them when the link returns. The simulation showed no lost alerts, but the Common Operating Picture can still become stale during an outage. These are software simulations, not proof that a physical military radio network will behave identically.”

Zero-egress air-gap check

- Reported audit: **0 outbound non-loopback sockets**, **0 external DNS attempts** and **0 external CDN/tile references** found in a 22-file code scan.
- 40 loopback sockets were monitored; one deliberately triggered socket-guard violation was blocked.
- Verdict in the source: **NO_EGRESS_OBSERVED**.
- Offline basemap tiles are generated procedurally on localhost; the prototype does not bundle a complete licensed worldwide MBTiles set.
- Scope limitation: application-level socket hooks and psutil monitoring were used. Kernel eBPF tracing and physical hardware network-tap capture are NOT DONE.

Judge explanation: “Air-gapped here means the audited application test observed no external network egress. We should not overstate the boundary: this is a user-space audit, not a hardware-certified air-gap assessment. A stronger deployment audit would add kernel-level tracing and physical network capture.”

KPI 11 — Multi-temporal satellite change detection

- Purpose: compare two image passes and flag objects that appear, disappear or move after aligning the scenes with ORB + RANSAC.
- Evaluation: 40 synthetic image pairs based on held-out tiles; 120 injected object edits (one new, one removed and one moved per pair), plus structural changes.
- Raw pooled results: precision 20.93%, recall 37.50%, F1 0.2687.
- Stability-filtered pooled results: **precision 80.95%**, recall 14.17%, F1 0.2411. The filter reduced false positives from 170 to 4 (**97.6% reduction**) but also missed many changes.
- Mean co-registration error across simulated 0–8 pixel shifts: **0.1449 pixels**. Structural anomaly capture: 65% of 40 injected examples.
- Real multi-pass satellite sea/land trials: NOT DONE.

Judge explanation: “The system aligns two images and compares detected objects to find possible changes. The stability filter removes many spurious changes, improving precision, but recall drops sharply. This is a clear trade-off: fewer false alarms, but more real changes missed. The reported tests use synthetic edits, so real satellite passes are the next validation step.”

Architecture and platform capabilities at a glance

- **Multi-Modal Sensors:** Sentinel-1 C-band SAR, xView optical imagery, drone EO/IR downlinks, unattended ground sensors (UGS) and AIS streams.
- **Dual-Engine AI:** YOLO11m generalist (optical) + YOLO11n 1024px vessel specialist fused via Weighted Box Fusion.
- **3 Distinct Offline Map Styles:** Satellite (dark navy ocean #0a1628, muted land, coastline glow), Dark Tactical (near-black, cyan graticule), Terrain (slate-blue ocean, synthetic elevation shading, watermark) + permanent tactical place names and maritime regions overlay.
- **Mission Planner OODA Interactivity:** Map zoom controls (+/- buttons, wheel, pinch), selectable intercept waypoints with map pan/highlight, animated dispatch fly-to (map.flyTo, zoom 8-9, 1.5s), marker pulse, and operator toast notifications.
- **Local Intelligence & Security:** Local BM25 RoE doctrine retrieval, STANAG 2014 SITREP compiler, PBKDF2-200k + JWT authentication, Docker containerized deployment.

Limitations judges may ask about

- Vessel detection is weak compared with aircraft and other classes; specialist improves from 12.75% to 16.24% but remains limited due to small 12–35px sizes and class imbalance.
- Dark-vessel, tracking, DDIL and change-detection figures are simulated rather than physical field trials.
- Jetson performance and power are unmeasured dense-FP16 roofline projections, not physical hardware benchmarks.
- Geolocation uses dataset GeoTIFF georeferencing, not independent differential GPS.
- Workload and mission-cycle reductions depend partly on assumptions; no human analyst or military-staff study has been completed.
- Procedural basemap supports offline operation but is not a complete licensed worldwide map dataset.

Suggested 3-minute demo sequence

- 1. **Joint COP Map:** Switch between Satellite, Dark Tactical, and Terrain offline styles; show tactical place names and maritime regions.
- 2. **Optical Detection:** Run detection and show the difference between generalist and vessel-specialist results.
- 3. **Dark-Vessel Intercept:** Open dark-vessel scenario; explicitly label as simulated and show AIS correlation/dead-reckoning logic.
- 4. **Analyst Workload Triage:** Show alert consolidation, auto-closed civilian vehicles, priority queue, and 0% missed critical threats.
- 5. **DDIL Blackout & Resync:** Trigger a simulated link outage, generate alerts, restore connectivity, and show zero-loss resynchronisation.
- 6. **Change Detection:** Show a two-pass change-detection example and explain the precision/recall trade-off.
- 7. **Mission Planner Dispatch:** Select an intercept waypoint, click “DISPATCH INTERCEPT MISSION” to trigger fly-to animation, compile STANAG SITREP, and show zero-egress audit proof.

Likely judge questions — evidence-based answers

Q: Is it production-ready?

A: No. It is a functional prototype with measured software benchmarks and simulated operational scenarios. Physical sea, radar, radio, GPS and human-staff trials remain.

Q: Why is vessel recall low?

A: The source identifies small 12–35 px boxes and class imbalance (280 vessels versus 42,824 infrastructure instances). A specialist model improves recall from 14.65% to 23.57% and mAP from 12.75% to 16.24%. The split audit proved zero data leakage across train, val_tune, and val_report holdout splits.

Q: Why do three different vessel mAP numbers appear in reports?

A: They evaluate different scopes and precision modes: 12.75% is the full 540-tile holdout (FP16), 12.07% is the 21-tile maritime holdout in edge_bench, and 12.26% is the 21-tile boost study baseline. Both 12.75% (generalist) and 16.24% (specialist) remain the authoritative headline values.

Q: What are the Jetson figures based on?

A: Dense-FP16 roofline scaling from our measured 35.74 ms power-capped host baseline: AGX Orin 44.7–48.8 ms (20.5–22.4 FPS), Orin Nano 134.6–203.1 ms (4.9–7.4 FPS). They are explicitly labeled 'unmeasured, rough estimate only'. TensorRT may improve this; unmeasured, no Jetson run.

Q: Can you guarantee zero network traffic?

A: The test observed zero outbound non-loopback sockets and zero external DNS attempts under user-space socket monitoring. Kernel eBPF and hardware packet capture were not done, so we claim “no egress observed in this audit,” not a universal guarantee.

Q: Are the latency claims end-to-end?

A: No. Tile inference (17.85 ms) and full-scene ingestion (mean 1.62 s, P95 2.30 s) are measured separately. The <1.8 s full decision-pipeline target is projected.

Q: Why are some workload reduction numbers negative?

A: Under Baseline A (manual raw review), Rakshak saves 86.54% review items. Under Baseline B (an arbitrary 40 min/scene cap = 8.0h/hr), reviewing 595.6 items/hr at ≥60s requires 9.93h–39.71h, exceeding a single analyst's 8.0h fixed budget (producing negative capacity differentials: -24.1%, -148.1%, -396.4%).

Q: What is next?

A: Prioritize real maritime-labelled data, physical Jetson hardware profiling, live sea/radar trials, independent GPS checks, and a controlled analyst workflow study.

Final submission / judging checklist

- Bring the raw JSON reports in evaluation/results/ that correspond to every KPI and make sure values shown match the report.
- Label every metric as measured, simulated, computed, assumed, projected or not done.
- Keep the vessel limitation, zero-leakage split audit, and change-detection precision/recall trade-off visible.
- State Jetson dense-FP16 roofline intervals (AGX Orin 44.7–48.8 ms, Orin Nano 134.6–203.1 ms) and label them 'unmeasured, rough estimate only'.
- Explain exactly what the zero-egress audit covers (user-space socket hooks) and what it does not cover.
- Separate item-count reduction (86.54%) from assumed analyst-time savings and explain Baseline B negative rows.
- Demonstrate the 3 distinct offline map styles, permanent tactical place names, and interactive Mission Planner dispatch fly-to.
- State that 12/12 smoke tests and 38/38 pytest tests pass, and code is synced on official repo HackFest-2026/Team1A.2.git.

Source note: *This guide was compiled directly from canonical evaluation JSON reports in evaluation/results/, evaluation/RESULTS.md, and PROJECT_OVERVIEW.md. All KPI values, test scopes, hardware platforms, caveats, and reconciliations are grounded in actual executed test scripts.*